PDF HTML5 PAGE FLIP

Next-Gen 3D Digital Flipbook Engine

Open-Source Synthetic Showcase Document

This document is purely generated for rendering and performance testing.

Contains zero private or customer-specific information.

Built with Vite, TypeScript, PDF.js, and HTML5 Canvas.

60 FPS HARDWARE ACCEL

1. Architecture & Rendering Engine

HIGH PERFORMANCE CLIENT-SIDE PDF RENDERING

Off-Main-Thread Processing with Web Workers

Mozilla PDF.js decodes PDF binary streams inside a dedicated worker, preventing UI thread freezes during intensive document rasterization.

Rendering Pipeline Highlights:

- HiDPI & Retina display automatic viewport devicePixelRatio scaling
- Lazy virtualization rendering: only visible pages render to canvas
- Active RenderTask cancellation on rapid user page flipping
- 3D curved spine shadow and realistic paper curl simulation

2. Performance Benchmarks

COMPUTATIONAL EFFICIENCY ON DESKTOP AND MOBILE

Feature / Capability	Legacy Turn.js	Page Flip Pro v2
Touch & Swipe Support	Limited (jQuery)	**Native Touch Events**
Animation Easing & FPS	~30-45 FPS	**Solid 60 FPS (GPU)**
Audio Synthesis	Audio Element	**Web Audio API**

3. React Component Usage

ZERO-CONFIGURATION MODERN FRONTEND WORKFLOW

```
// Install package from npm
npm install pdf-html5-page-flip

// Import in your Next.js / React application
import { PdfFlipbook } from 'pdf-html5-page-flip/react';
import 'pdf-html5-page-flip/style.css';

export default function Reader() {
    return <PdfFlipbook src="/my-book.pdf" theme="dark" />;
}
```

4. Interactive Controls

INTUITIVE NAVIGATION AND READING TOOLS

Keyboard Shortcuts

Left / Right Arrows : Prev / Next

Home / End : First / Last Page

F Key : Toggle Fullscreen

Sidebar & Tools

Outline : Hierarchical TOC

Search : Full-text query

Thumbnails : Visual preview

Ready to Embed in Your App

Deploy with zero dependencies on modern browser standards.

License: Apache License 2.0

Repository: github.com/iamapinan/pdf-html5-page-flip

Live Demo: https://flipbook.apps.gracer.ai